

§1

THE SITUATION

Ukraine solved the arithmetic that was bankrupting air defence. Their Sting interceptor costs two thousand dollars against the hundreds of thousands or millions a missile costs, and it has taken down several thousand drones. We cannot have it. Every one of those interceptors is flown by a trained pilot in goggles, one pilot per interceptor per engagement, sustained by a mobilised wartime population we do not have — and Ukraine has banned the export of them regardless. Meanwhile the target has outrun the answer: the jet drones now in mass production cruise at 300 to 400 km/h and the newest reach 500 to 600, faster than the interceptors chasing them. Even a copy would arrive obsolete, and building our own stock is the only route left.

PROBLEM STATEMENT

Ideate one layer of an autonomous interceptor swarm that needs no pilot — the drone itself, the terminal guidance, the swarm deployment logic, or the patrol and recharge cycle that keeps it airborne. Build it in software and showcase it working, measured against a stated baseline.

§2

THE FOUR LAYERS

Select one. Assume the interceptor runs the target down from behind rather than meeting it head-on. Nothing here can be flown in Singapore, so every deliverable is built and tested in software, with hardware wherever hardware makes the result believable.

01

DRONE AND PROPULSION DESIGN

To run down a jet drone cruising at 500 to 600 km/h you have to fly faster than it does, and survive both the acceleration to get there and the heat of holding it. Design the entire aircraft in software — airframe, propulsion, power, thermal management, control surfaces, structure — not a chassis. Test it in simulation and report time-to-speed, sustained top speed, thermal limit and structural margin in a high-G turn. Then 3D print it and bring the article, so the design can be seen as well as read.

02

TERMINAL GUIDANCE AND THE SEEKER

Overtaking from behind, your closing rate is only the difference between your speed and the target's, so the engagement is decided in the last few hundred metres — which is exactly where the target jinks, and where a decoy or a bird will cost you the round. You have no satellite fix, no tether and no pilot. Build the seeker and guidance stack — sensor, compute and control loop — to a stated end-to-end latency budget. Run it against a target executing structured evasion, report hit rate and miss distance against a

proportional-navigation baseline, and demonstrate it rejecting a decoy or a bird and aborting the run.

03

ONBOARD AUTONOMY AND SWARM DEPLOYMENT

A raid arrives as a swarm and the answer has to be a swarm. Two decisions sit here: which of your interceptors are mobilised at all — each has a cost, a position, and a state, airborne on patrol or docked on the ground — and then which of your drones takes which of theirs. Both have to be agreed between the drones themselves, because the ground link is the first thing an adversary removes. Build the coordination and allocation engine and run it against a simulated incoming swarm. Report leakage against a naive one-round-per-target baseline, and show drone-to-drone coordination, target reassignment when a round fails, and collision avoidance between your own aircraft.

04

AUTONOMOUS PATROL AND SUSTAINMENT

An interceptor is only useful if it is airborne when the raid arrives, and a battery lasts minutes. Holding a patrol overhead continuously means a rotation — on station, transiting home depleted, charging, ready to launch — running with nobody present. Build a model of that cycle covering fleet size, endurance, transit time, charge time and dock capacity, and show what coverage it sustains over a continuous period. Design the autonomous docking and recharging station that makes it possible, and report the fleet size required to keep one interceptor permanently on station.

§3

THINGS TO TAKE NOTE OF

01

LOW LATENCY IS CRITICAL

The terminal phase is decided in the last few hundred metres, and a jinking target changes the required correction faster than a human could issue it. Every millisecond between sensing a manoeuvre and acting on it becomes miss distance. That is why the pilot has to go, and why the latency budget is a headline number rather than an implementation detail.

02

UNIT COST IS SET BY THE SEEKER

A fifteen-thousand-dollar seeker on a two-thousand-dollar airframe is a seventeen-thousand-dollar round. Any design that assumes the guidance is free has not solved the economics it claims to.

03

COLLATERAL DAMAGE OVER A DENSE CITY

There is nowhere here that is nowhere. Net and entanglement capture exists precisely to limit debris, and over this ground that moves from preference towards requirement.

04

DEGRADED CONDITIONS DRIVE THE LEAKAGE

Roughly one attacking drone in ten still arrives, and most arrive at night or in rain. That is the finding failing, not the interception — which makes the cue chain, not the airframe, the usual limiting factor.

05

ALLOCATION IS AN ECONOMIC INSTRUMENT

Cheap rounds against slow targets, expensive rounds held for fast ones. Expending a jet interceptor on a decoy loses the exchange even when the interception succeeds, which makes the mobilisation decision in layer three a matter of cost as much as of geometry.